

Defence presentation outline

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1. Content (1 min)

- (a) Intro, chapter 1, chapter 2, conclusion; will last 15-20 minutes.

2. Introduction

- (a) ACC lengthens growing seasons
 - i. Happens through earlier SOS + delayed EOS
 - ii. (Figure): conceptual figure simplified showing SOS and EOS shifts
- (b) GS are longer, but also warmer
 - i. Total annual GDD increases
 - ii. (Figure): conceptual figure with GDD accumulation curve
- (c) SOS + EOS shifts = more GDD, but shifts are asymmetric
 - i. Impacts how much GDD is accumulated in the GS
 - ii. SOS shift more, but are cooler
 - iii. EOS shift less, but are warmer
 - iv. (Figure): conceptual figure with showing how much GDD in each season
- (d) SOS + EOS shifts = longer and warmer seasons; should increase growth
 - i. Recent studies failed to find this relationship
 - ii. Important because temperate forest = 40% of global net carbon sink
 - iii. (Figure): TBD
- (e) Question: do trees grow more with longer and warmer seasons?
 - i. Objectives: address GS/growth disconnection using two systems
 - ii. Chapter 1: juvenile trees, 4 species, 4 prov, leaf phenology and tree rings
 - iii. Chapter 2: mature trees, 11 species, leaf phenology and tree rings
- (f) Recent research showed that this may not be the case
 - i. 1
- (g) Drought and competition as potential growth inhibitors and internal constraints
 - i. 1
- (h) Questions: do longer and warmer seasons increase growth in juvenile and mature trees?
 - i. 1
- (i) Objectives: answer this question using north american juvenile trees of 4 species from 4 provenances (chapter 1) and mature trees of 11 species (chapter 2)
 - i. 1

3. Chapter 1

- (a) Introduction: gs conceptual figure

- (b) Introduction: constraints on growth
 - (c) Methods: trees grown from seeds and sourced from 4 provenances; 3.5 degree lat gradient; phenology for 3 years, growth with tree rings
 - (d) Methods: bayesian hierarchical models? 4 different predictors
 - (e) R/D: More species grew more with warmer seasons than calendar season; because warmer is a better predictor of growth
 - (f) R/D: Later EOS increased growth more than SOS; because higher temperature than SOS; suggests looking more at EOS
 - (g) R/D: No provenance effect; at least for this small gradient, no evidence of local adaptation on growth
 - (h) Conclusion: juvenile trees should grow more, means it would be good for range expansion and forest regeneration
4. Chapter 2
- (a) Introduction: different growth strategies could influence tree growth responses to GS length
 - (b) Introduction: fast vs slower growing species
 - (c) Introduction: wood porousness and timing of leaf phenology and growth = differences in temperature experienced; also correlates with different life-history strategies
 - (d) Introduction: carbon investment in storage may blur annual growth responses
 - (e) Methods: citizen science leaf phenology; growth with tree rings at the arboretum
 - (f) Methods: stats? Same predictors, but used leaf coloring as EOS.
 - (g) R/D: very little effect of season length
5. Conclusion
- (a) 1
 - (b) 2